The World Wide Web

The World Wide Web is a network of information that is accessed via the Internet (see pp.196–197). It consists of interlinked web pages formatted in a common language and identified by unique addresses.

How the World Wide Web works

The Web is a vast network of multimedia pages, navigated and downloaded using a program called a browser. Web pages are interlinked. A collection of linked and related web pages with a common domain name comprises a website. Each web page is identified by a unique uniform resource locator (URL), which specifies its location. Browsers retrieve these pages from servers as documents formatted using hypertext markup language (HTML) and render them as readable multimedia pages. The hypertext transfer protocol (HTTP) sets out the procedures for communication between browsers and servers on the World Wide Web.



1 User searches
A user accessing a search engine types one or more keywords relevant to his query before clicking on a search button or hitting "enter" to start the search process.

2 Request
The search terms are sent to the wider Internet via a router. They are directed to the search engine's servers.

Router connects user to wider Internet

ROUTER

Searches are processed by data centers running many powerful computers

DATA CENTER

3 Search index
A computer scans the search engine's index for the most relevant and reputable pages containing those search terms.

Searching the Web
Web pages are often accessed using programs called search engines rather than entering URLs directly. Search engines crawl through web pages to create an index, which is used to generate search results. These results are presented as a list of relevant links.

HTML

HTML is a language used to design web pages. Browsers receive HTML documents from web servers and render them into readable web pages containing text and other media. Codes called HTML tags are used to add and structure content in the page; for example, <img> introduces an image, while <a> inserts hyperlinks to web pages, files, or email addresses.

```
<!DOCTYPE HTML>
<HTML>
<BODY> </BODY>
</HTML>
```

Internet protocols

Hypertext transfer protocol (HTTP) is the universal set of rules underlying use of the World Wide Web. HTTP forms the basis for processing web documents and how servers, browsers, and other agents respond to commands. When a user enters a URL to access a web page, his browser looks for the web server's Internet address using the domain name system (DNS). It then sends a request to a web server, which issues a response with a status code containing information, such as whether the URL is valid so that the page can be loaded. A sequence of requests and responses is called an HTTP session.

HTTPS
Hypertext transfer protocol secure (HTTPS) uses the encryption protocol transport layer security (TLS). This gives the user privacy and security when browsing online.

